Ashvin Anilkumar

Madison, WI | anilkumar4@wisc.edu | (608) 395-4238 | linkedin.com/in/ashvin-anilkumar | github.com/ashvin-a

ashvin-anilkumar.vercel.app

Summary

Graduate ECE student at UW-Madison specializing in **Robot Learning** with a special interest in **Reinforcement Learning** for Robotic applications with **2 years** of work experience building scalable AI products. Proven experience deploying scalable AI architectures and deterministic C++/ROS 2 pipelines for physical robot autonomy.

Education

University of Wisconsin-Madison, Master of Science in Electrical and Computer Engineering — December 2026
- *Relevant Coursework:* Computer Vision, Topics in Advanced Robotics, High-performance Computing, Optimization.

Government Engineering College, Barton Hill, Trivandrum, Bachelor of Technology in Electrical and Electronics Engineering — July 2023
- *Additional Coursework:* Minor in Information Technology

Technical Skills

Core Robotics & Autonomy: ROS 2, SLAM, Motion Planning, Kinematics, Sensor Fusion, Behavior Trees, Sim-to-Real

Embedded Systems & Hardware: I2C, SPI, UART, STM32, Raspberry Pi Pico, Motor Drivers/Controllers, PCB Design & Assembly, Embedded C, Kalman Filter

Computer Vision & Perception: OpenCV, PCL, Visual Odometry, Multi-sensor Fusion, Semantic Segmentation, Pose Estimation

AI & Machine Learning: PyTorch, JAX, Reinforcement Learning, RLHF, State Space Models, Transformers

Systems, Simulation & Tools: Linux, Docker, Kubernetes, CMake, Eigen, Gazebo, Isaac Sim, MuJoCo, Git, Claude Code

Programming: C++ (14/17), Python, CUDA, Rust, Bash

Publications

Chrono Agentic: Evidence-Grounded Agents for Executable World Simulation — June 2026

arXiv:2605.14398 (under review, ICLR 2027)

Hongyu Wang, Jingquan Wang, **Ashvin Anilkumar**, Bocheng Zou, Radu Serban, Dan Negrut

- Spearheaded a multi-agent system turning natural-language specifications into executable **PyChrono** simulations; **82.5%** of generated demos met semantic and physical correctness on PhyWorldBench vs. **52.5%** for text-to-video baselines.

Informationally Decoupled Trajectory Design for Sim-to-Real System Identification — Sep 2026

arXiv:2610.10905 (under review, IEEE International Conference on Robotics and Automation (ICRA) 2027)

Sangwoo Shin, **Ashvin Anilkumar**, Ryan Gao, Josiah Hanna

- Introduced IDTD, an exploration objective for sampling-based system identification built on the Fisher-information Schur complement score, so collected trajectories expose separately identifiable parameters instead of ambiguous combinations.
- Reduced parameter identification error by **39.6%** on average over the strongest active-exploration baseline across Go2, G1, and Crazyflie simulations, improved downstream policy transfer, and validated on a real **K1 humanoid**.

Reinforcement Learning from Real-Robot Interaction without High-Quality State Estimates — Sep 2026

arXiv:2610.14042 (under review, IEEE International Conference on Robotics and Automation (ICRA) 2027)

Ryan Gao*, **Ashvin Anilkumar***, Nicholas Corrado, William Cong, Josiah Hanna — **Equal contribution*

- Introduced RSSM-RLPD, which learns a recurrent state-space representation corrected only when the ball is observed, enabling a NAO robot to learn soccer goal-scoring from on-board UKF state estimates with no external instrumentation.
- Reached **100%** success from multiple starting positions after **30 minutes** of real-world interaction vs. **0%** for a feed-forward policy, and outperformed behavior cloning, feed-forward, and GRU baselines in simulation.

Work Experience

Graduate Research Assistant, University of Wisconsin - Madison, United States — Dec 2025 – Present

Reinforcement Learning | *MuJoCo* | *ROS 2* | *JAX* | *Sim-to-Real* | *Perception*

- Architected sample-efficient RL pipelines for continuous control on physical hardware; advised by **Prof. Josiah Hanna**.
- Engineered deterministic agentic workflows in **PyChrono** to automate simulation setups; advised by **Prof. Dan Negrut**.
- Designed and deployed reactive behavior trees on **Booster K1** hardware, helping the team place **4th** at RoboCup Asia-Pacific.

AI Engineer, Backend Development (AI/ML), Qburst Technologies - Trivandrum, India — Aug 2023 – Aug 2025

Project: Scalable AI Search

FastAPI | *LangChain* | *LlamaIndex* | *Postgres* | *MongoDB* | *Hugging Face*

- Developed an AI search bar (**FastAPI**) integrating **RAG** and natural-language-to-SQL via **LangChain** & **LlamaIndex**.
- Deployed cloud-native microservices using **FastAPI** and Python to handle complex global influence data.
- Built optimized text embeddings in **Postgres** and hosted few-shot examples in **MongoDB**.
- Implemented agentic RAG workflows, optimizing token usage by **56%** and enhancing system response speed.
- Increased the accuracy of search results by **90%** and expanded the range of questions it can handle by **85%**.
- Practiced **CI/CD** in an Agile environment, driving software from R&D to production with cross-disciplinary teams.

Project: AI Report Reviewer System
FastAPI | LangChain | Azure

- Designed the architecture for the project, making the system extremely deterministic and consistent in generating ratings.
- Formulated an AI system for automated evaluation and structured feedback on reports using **LangChain** and **FastAPI**.
- Achieved **89%** alignment with human ratings, improved evaluation accuracy by **35%** and reduced review costs by **67%**.

Research & Academic Projects

RoboCup 2026 Dec 2025 – July 2026
Isaac Sim | MuJoCo | ROS 2 | PyTorch | Rerun

- Engineered a decentralized multi-agent coordination stack using **ROS 2** and deep reinforcement learning for a fleet of autonomous **Booster K1**s.
- Migrated simulation environments from **Isaac Sim (PhysX)** to **MuJoCo**, enhancing contact dynamics fidelity and boosting training reliability by **40%**.
- Developed the localization algorithm to optimize agent movement in high-density, dynamic environments.
- Bridged the sim-to-real gap by developing a custom deployment pipeline, reducing iterative hardware testing overhead by **20%**.
- Placed **4th** at RoboCup Asia-Pacific (Middle Division), held at the **World Humanoid Robot Games 2026**, Beijing.

Performance Analysis of Obstacle Avoidance algorithm Mar 2026 – Apr 2026
CUDA | OpenMP | Perf

- Architected a deterministic obstacle avoidance pipeline utilizing **HOG** feature extraction and SVM classification.
- Profiled throughput and latency bottlenecks, successfully accelerating execution via OpenMP multi-threading and custom CUDA kernels, achieving **8.4x** and **33.5x** speedups from the baseline, respectively.

Real time Visual Odometry using State Space Models Mar 2026 – Apr 2026
State Space Models | PyTorch | ROS 2 | Gazebo

- Benchmarked **State Space Models** (SSMs) for real-time visual odometry inference under strict compute constraints within a ROS/Gazebo environment.
- Developed a simulation of a differential drive robot equipped with a monocular RGB camera in Gazebo.
- Synthesized multi-modal datasets of sequential imagery and ground-truth trajectories via precise teleoperation of a differential drive robot.

Learn RL - A Reinforcement Learning Tutorial Jan 2026 – Present
MuJoCo | ROS 2 | PyTorch | OpenAI Gym | Stable Baseline 3

- Open-source tutorial covering foundational reinforcement learning concepts with hands-on implementations, including PPO and SAC algorithms using **OpenAI Gym** and **MuJoCo** physics engines.
- Designed progressive exercises guiding learners from tabular methods to continuous control with **Stable Baseline 3**.

University Rover Challenge: Perception & Autonomous Navigation Sep 2025 – Dec 2025
ROS 2 | Gazebo | DepthAI | RViz| RANSAC

- Built a real-time **perception and mapping** stack for autonomous navigation on unstructured terrain in **ROS 2 Humble**.
- Developed a ground detection algorithm using **RANSAC** and improved the system reliability by **30%**.
- Utilized **DepthAI** and **PyTorch** to deploy semantic segmentation models, enabling the robot to distinguish between navigable terrain and obstacles.
- Implemented **Path Planning** and **Movement** controls using ROS 2, fusing stereo vision (DepthAI) and IMU data.

CubeSat Attitude Determination & Control System (ADCS) May 2022 – May 2023
Embedded C | STM32 | Raspberry Pi Pico | I2C | UART | Kalman Filter

- Fabricated a full ADCS stack for a CubeSat capstone project, pairing a **Raspberry Pi Pico** flight computer with an **STM32 Discovery** motor controller over **UART**, interfacing a **BMI160** IMU over **I2C** and driving custom soft-iron-core magnetorquers.
- Applied a hybrid **Bang-Bang/B-dot** control law for detumbling and the **TRIAD** algorithm for three-axis attitude determination from IMU and magnetometer readings.
- Regulated sensor noise and drift using a **Kalman filter** for onboard state estimation and calibration.